

Multisystem Inflammatory Syndrome in Children (MIS-C): Care Guide for Children and Adolescents in Alberta

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Multisystem Inflammatory Syndrome in Children (MIS-C) Key Messages

- MIS-C is a rare complication of SARS-CoV-2 infection
- Consider MIS-C in children or adolescents with any of the following:
 - Fever for 3 or more days, with a recent history of SARS-CoV-2 infection OR known exposure
 - Fever accompanied by abdominal pain or significant vomiting/diarrhea
 - Fever accompanied by shock
 - Signs and symptoms similar to Kawasaki Disease and Toxic Shock Syndrome (TSS)
- Do not delay anti-microbial therapy because you suspect MIS-C, as it is a diagnosis of exclusion

For physicians working at Alberta Level 1 and 2 Emergency Departments - if you suspect a child may have Multi-System Hyperinflammatory Syndrome (MIS-C) - please contact RAAPID North or RAAPID South and ask to speak (for a stable patient) to a Children's Hospital ED physician or (for an unstable patient) a Pediatric Intensivist. Refer to the MIS-C case identification poster on page 10.

Introduction

This document addresses Multisystem Inflammatory Syndrome in Children (MIS-C), a condition characterized by fever, inflammation, and multiorgan dysfunction that manifests late in the course of what is often an asymptomatic or mild SARS-CoV-2 infection.¹ These patients present with an acute febrile illness accompanied by a hyper-inflammatory syndrome and often suffer multi-organ failure, particularly acute heart failure, some weeks after an acute COVID-19 infection or exposure to COVID-19.¹ Early recognition and appropriate transfer and management of these patients is critical. Children may present with shock and may not tolerate large volume fluid resuscitation. These patients should be managed at a tertiary pediatric centre. Although these children are often critically ill, most children ultimately will have favorable outcomes. Comprehensive data for Canada is not available but data for the US is available through the Center for Disease Control; as of Oct 31st, 2022, 9,073 children have met CDC's case definition for MIS-C, and 74 have died (0.8%).²

Highlights from the CDC data include:

- The median age of children with MIS-C is 9 years, with half between 5 and 13 years of age.
- Of children whose race/ethnicity is known (8,557), 56% are either Hispanic/Latino (2,235 cases) or Non-Hispanic Black (2,595 cases).
- 98% of children had a positive test result for SARS CoV-2; the remaining 2% had contact with someone with COVID-19.
- 60% of children are male.²

MIS-C Case Definition

For this guideline we have adapted the case definition for MIS-C is that used by the U.S. Centers for Disease Control and Prevention (with a modification to fever duration).³ ***Clinicians should use the Diagnostic Pathway (next page) to establish a diagnosis of MIS-C rather than this Case Definition. Currently the accuracy (i.e. sensitivity/specificity) of various case definitions and diagnostic tests have not been established.***

- An individual aged <21 years presenting with fever*, laboratory evidence of inflammation**, and evidence of clinically severe illness requiring hospitalization, with multisystem (>2) organ involvement (cardiac, renal, respiratory, hematologic, gastrointestinal, dermatologic or neurological); AND
- No alternative plausible diagnoses; AND
- Positive for current or recent SARS-CoV-2 infection by RT-PCR, serology, or antigen test; or exposure to a suspected or confirmed COVID-19 case within the 4 weeks prior to the onset of symptoms.

*Fever >38.0°C for ≥72 hours [Note for our guideline we have modified duration of fever to be consistent with other parts of this guideline.]

**Including, but not limited to, one or more of the following: an elevated C-reactive protein (CRP), erythrocyte sedimentation rate (ESR), fibrinogen, procalcitonin, d-dimer, ferritin, lactic acid dehydrogenase (LDH), or interleukin 6 (IL-6), elevated neutrophils, reduced lymphocytes and low albumin

Differential Diagnosis

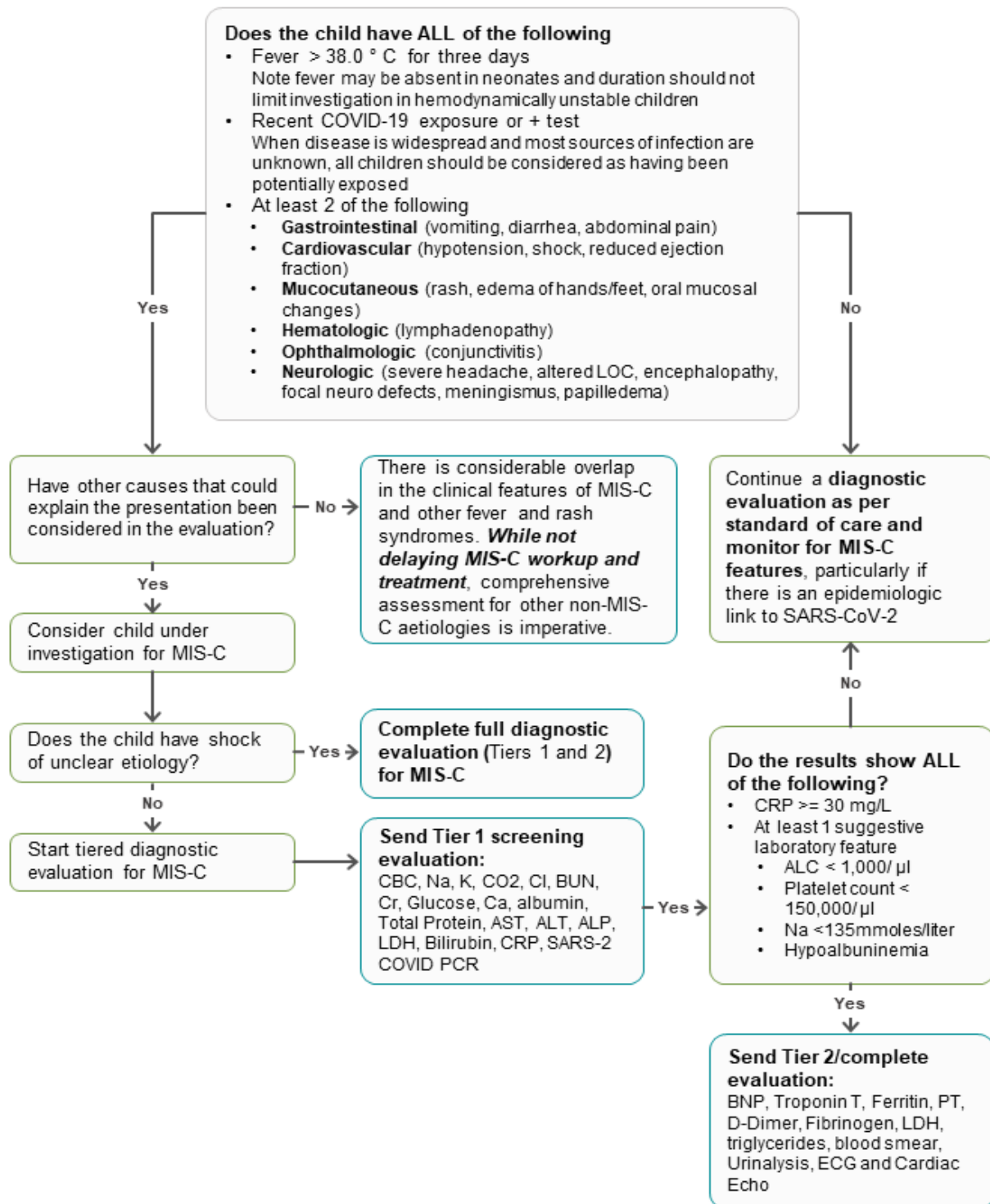
There is an overlap with various other inflammatory syndromes such as cytokine storm syndrome (CSS) seen in adult and pediatric COVID-19 ARDS, macrophage activating syndrome (MAS), hemophagocytic lymphohistiocytosis (HLH), systemic inflammatory response syndrome (SIRS), culture-negative sepsis, multi-organ dysfunction syndrome seen in septic shock, cytokine release syndrome (CRS) in oncology CAR-T therapy, and, as already mentioned, KD and TTS. MIS-C overlaps with KD with associated cardiogenic shock/myocarditis (KDSS). Coronary aneurysms have been described in MIS-C.¹

In MIS-C, gastrointestinal symptoms including severe abdominal pain are very common.¹ Cases have been mistakenly managed as surgical emergencies. As such, abdominal ultrasound (US) and/or CT scan may be recommended to rule out appendicitis, intussusception and other acute abdominal pathologies.

Key points that help distinguish MIS-C from KD unrelated to SARS-CoV-2 include:

- MIS-C patients are more commonly of African, Afro-Caribbean, or Hispanic descent, but less likely to be of East Asian descent
- MIS-C patients encompass a broader age range, have more prominent GI and neurologic symptoms, present more frequently in shock, and are more likely to display cardiac dysfunction (arrhythmias and ventricular dysfunction) than children with KD
- At presentation, MIS-C patients tend to have lower platelet counts, lower absolute lymphocyte counts, and higher CRP levels than patients with KD.
- Epidemiologic studies of MIS-C suggest that younger children are more likely to present with KD-like features while older children are more likely to develop myocarditis and shock
- It is unknown if the incidence of coronary artery aneurysms (CAA) is different in MIS-C compared to KD; however, MIS-C patients without KD features can develop CAA¹

Diagnostic Pathway



Management Algorithm

Admitted to inpatient unit - no need for vasopressors - no cardiac dysfunction Treatment - IVIG 2gr/kg (max 70 gr)¹ plus - IV Methylprednisolone or PO equivalent 2 mg/kg (max 60 mg) x 5 days then (MP taper) 1 mg/kg x 5 days then 0.5 mg/kg x 5 days - Low dose ASA 3 to 5 mg/kg/day, max dose 81mg² - Consider LMWH/heparin if coronary aneurysm, reduced cardiac ejection fraction, or documented thrombosis³	Admitted to PICU - hemodynamic instability requiring vasopressors - no cardiac dysfunction Treatment - IVIG 2gr/kg (max 70 gr)¹ plus - Methylprednisolone IV 30 mg/kg/day (max 1g) x minimum 3 days or at least 24 h off vasopressors (whichever longer) then IMP taper) - Methylprednisolone IV 10 mg/kg/day x 2 days then step down to - IV methylprednisolone or PO prednisone 2mg/kg/day (max 60 mg) x 5 days then 1 mg/kg/day x 5 days then 0.5 mg/kg/day x 5 days - Low dose ASA 3 to 5 mg/kg/day, max dose 81mg² - Consider LMWH/heparin if coronary aneurysm, reduced cardiac ejection fraction, or documented thrombosis³	Admitted to PICU - hemodynamic instability requiring vasopressors - cardiac dysfunction (reduced ejection fraction) Treatment - IVIG 2gr/kg (max 70 gr)¹ plus - Methylprednisolone IV 30 mg/kg/day (max 1g) x minimum 3 days or at least 24 h off vasopressors (whichever longer) plus - Anakinra max 10 mg/kg/day iv divided QID then (MP taper) - Methylprednisolone IV 10 mg/kg/day x 2 days then - Methylprednisolone IV 5 mg/kg/day x 2 days then step down to - IV methylprednisolone or PO prednisone 2mg/kg/day (max 60 mg) x 7 days then 1 mg/kg/day x 7 days then 0.5 mg/kg/day x 7 days - Low dose ASA 3 to 5 mg/kg/day, max dose 81mg² - Consider LMWH/heparin if coronary aneurysm, reduced cardiac ejection fraction, or documented thrombosis³
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Steroid taper highlighted in grey

Refractory disease defined as persistent fever and significant end-organ involvement?



- Methylprednisolone IV 30 mg/kg/day (max 1g) x minimum 3 days
- Consider Anakinra or Infliximab in consultation with Rheumatology

- 1 Consider consulting Rheumatology for any patient with diagnostic or therapeutic uncertainty.
- 2 Do not treat with ASA if the patient has significant risk of bleeding
- 3 At Stollery, consult KidsCLOT regarding the need for LWMH/heparin for all children diagnosed with MIS-C.
- 3 At Alberta Children's consult cardiology for children with coronary aneurysm or decreased LV ejection fraction, or hematology for children with a documented thrombosis regarding the need for LWMH/Heparin.

For confirmed Kawasaki Diagnosis

- Follow Kawasaki protocol
- Consider Cardiology referral +/- Rheumatology
- IVIG 2 gram/kg. Max 70 grams
- ASA: moderate dose 30-50 mg /kg/day

Laboratory and Diagnostic Testing

Potential MIS-C Lab Investigations

Tier 1- MIS-C Symptoms without Hemodynamic Symptoms

☐ CRP	☐ Glucose
☐ CBC	☐ Ca
☐ Na	☐ BUN
☐ K	☐ Albumin
☐ CO2	☐ Total Protein, AST, ALT, ALP, Bilirubin
☐ Cl	☐ NP swab/ aspirate throat swab for respiratory panel plus SARS-CoV-2 PCR
☐ Cr	

Tier 2- MIS-C Symptoms with Hemodynamic Symptoms or Positive Tier 1 Labs: Perform all of Tier 1 level tests plus:

☐ Pro BNP*	☐ Triglycerides
☐ Troponin T*	☐ Blood smear
☐ Ferritin	☐ U/A
☐ PT	☐ SARS CoV-2 serology
☐ D-Dimer	☐ Echocardiogram
☐ Fibrinogen	☐ ECG q 48 hours
☐ LDH	

** See Appendix for NT-proBNP and Troponin T Normal Values in Children*

Treatment

Treatment should be guided by consultation (as indicated) with PICU, Infectious Disease, Rheumatology, Cardiology, and Hematology but should be performed with the following considerations in mind:

- Volume resuscitation—should be minimized due to cardiogenic nature of shock and significant capillary leak. We recommend 5 ml/kg aliquots of crystalloid and commencement of vasopressors (initially norepinephrine) once 20 mls/kg volume resuscitation is reached. Volume resuscitation should be ended before the 20 ml/kg volume if hypotension is not responsive to fluid, cardiac function is known to be depressed (e.g. POCUS assessment), and/or the CXR shows evidence of pulmonary edema/effusions
- If clinical concern for sepsis or other localized infection, obtain cultures and initiate antibiotics to cover for bacterial sepsis/TSS with ceftriaxone and clindamycin +/-

vancomycin

- Norepinephrine initially for warm shock should be started early rather than excessive use of IV fluids. Consider addition of epinephrine depending on clinical circumstances and cardiac ECHO results
- IVIG 2 grams/kg IV—give slowly if volume load is of concern. Max dose of 70 grams.
- Aspirin Low dose 3-5mg/kg/day, MAX 81mg
- Systemic steroids methyl-prednisone—In patients presenting with shock, early use of IV methylprednisolone and IVIG is associated with a more favorable course. If there is significant inflammation or cardiac involvement IV pulse methylprednisolone of 30 mg/kg/day (MAX 1 gram) can be given even prior to IVIG.

Other treatment options to consider (tuberculosis should typically be excluded before initiating these biological agents) – these agents should only be used following discussion with pediatric rheumatology:

- Anakinra—First line treatment for patients with refractory MIS-C and/or with persistent end organ involvement despite treatment with IVIG and systemic steroids.¹
- Infliximab—A secondary potential therapeutic agent in patients with refractory MIS-C. It should not be used any patients with potential Macrophage Activation Syndrome (MAS).¹

Other therapeutic considerations:

- Anti-acid treatments should be used to prevent gastrointestinal complications in MIS-C patients taking corticosteroids and ASA.¹
- PTE/DVT prophylaxis—patients with large coronary aneurysms and moderate to severe LV dysfunction should be treated with anticoagulants.¹ The use of anti-coagulants for other indications are uncertain.¹

Discharge and Follow-Up

Follow-up suggested (based on national guidance):

1. Cardiology follow-up with ECHO at 1-2 and 6 weeks after diagnosis for children with uncomplicated cases. For children with significant cardiac involvement, consult cardiology for specific recommendations. Of note, if dysfunction or coronary dilation/aneurysms are present, repeated echocardiography while in hospital should be performed to ensure plateauing of severity. At discharge, patients should be restricted from competitive sports and/or moderate-vigorous exercise until follow-up with pediatric cardiology has occurred. Investigations to be considered at these visits include (as appropriate) ECG, echocardiography, exercise stress testing, and MRI. If patients are treated with corticosteroids at discharge— a slow steroid taper over 2-4 weeks with close monitoring of symptoms and inflammatory markers is needed to avoid rebound inflammation.
2. Rheumatology follow up for patients requiring anakinra.
3. Pediatric follow-up after discharge. However the timing, location and type of follow-up varies across the province. Consult local hospital recommendations for details.
4. Return if febrile or unwell

Reporting

MIS-C is a notifiable disease in Alberta.

The current AH reporting requirements are outlined in the MIS-C Notifiable Disease Guideline:

[Public Health Disease Management Guidelines: Multisystem Inflammatory Syndrome in Children \(MIS-C\) \(alberta.ca\)](#)

*All physicians **shall** notify the Medical Officer of Health (MOH) (or designate) of the zone, of all cases that meet the preliminary case definition in the prescribed guideline above as determined by the MOH within 48 hours (two business days) and include:*

- ✓ Age,
- ✓ Date of birth,
- ✓ Gender,
- ✓ Personal health number,
- ✓ Positive laboratory report (if applicable)
- ✓ Other relevant clinical/epidemiological information.

***Report here: link to one page form**

<https://www.alberta.ca/assets/documents/health-ab-mis-children-case-report-form.docx>

For Edmonton Zone please report to MOH

Lead Dr Christopher Sikora

Phone: 780-433-3940

For Calgary Zone please report to MOH

Lead Dr Nicholas Etches

Phone : 403-955-3369

Research

Various research studies are underway in Alberta for COVID related illnesses:

Studies in Alberta- for more information contact Drs. Suzette Cooke, ACH or Joan Robinson, U of A
University of Calgary: <https://research.ucalgary.ca/covid-19/collaboration>

Canada: <https://cancovid.ca>

Parent Resource

A resource for parents provided by the *U.S. Center for Disease Control* can be accessed at this website:

<https://www.cdc.gov/mis/mis-c.html>

Appendices

Troponin T and NT-proBNP Normal Values (ng/L)

Table 1. Pediatric reference intervals and cut-offs for high-sensitivity cardiac troponin T (hs-cTnT) and N-terminal pro-B-type natriuretic protein (NT-proBNP).

Analyte	Age range	n	Lower limit (90% CI) ^a (ng/L)	Upper limit (90% CI) (ng/L)	99 th percentile Cut-off (90% CI) (ng/L)	% > 3 ng/L	% > 5 ng/L
hs-cTnT	0–<6 months	64	7 (4, 10)	78 (68, 87)	87 (76, 97)	100%	N/A
	6 months–<1 year	45	6 (5, 7)	34 (28, 42)	39 (32, 47)	100%	N/A
	1–<19 years	131	<3 (<3, <3)	9 (6, 11)	11 (11, 14)	30%	N/A
NT-proBNP	0–<1 year	211	39 (26, 57)	3569 (2188, 4876)	5272 (4788, 6937)	N/A	100%
	1–<19 years	265	<5 (<5, 7)	178 (144, 203)	216 (182, 250)	N/A	97%

^aCI, confidence interval.

Poster for ED and Urgent Care Centres

Multisystem Inflammatory Syndrome in Children (MIS-C) Associated with COVID

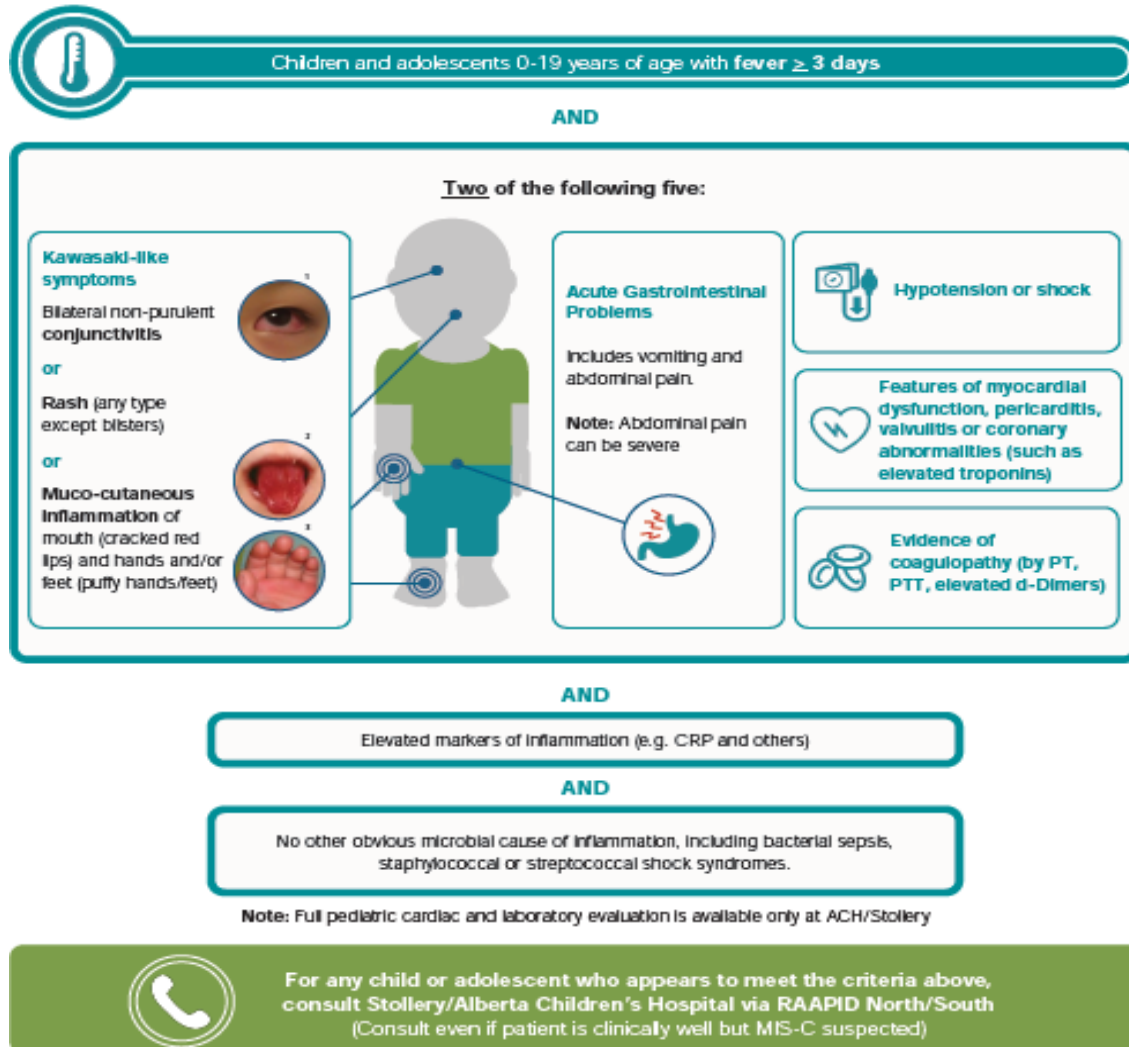
Children and adolescents can develop a severe inflammatory syndrome thought to follow exposure to COVID which has features similar to Kawasaki Disease and/or Toxic Shock Syndrome (TSS).

Vigilance for this uncommon but serious disease is critically important.

Children and adolescents can present in a wide variety of ways and ranging severity, including fever and shock.

Note: Patient may not have known COVID exposure or positive COVID testing.

Criteria of MIS-C:



While awaiting disposition decision, continue to provide supportive care and watch for adverse response to fluid resuscitation.
Do not delay anti-microbial therapy because of suspected MIS-C.

¹Pharid, C. (May 08, 2020) Kawasaki Disease From COVID-19 is Rare: How Common? Medpage Today. Retrieved from: <https://www.medpagetoday.com/infectiousdisease/covid19/56393>
²Wier D, Dack R, Kuo H-C. Mercury promotes catecholamines which potentiate myocardial autonomic and vasodilation: implications for isolated 1,4,5-triphosphate 3-kinase C susceptibility in Kawasaki syndrome. Korean Circ J. 2013;43(5):581-591.
³Rodriguez, T. (May 16, 2017) Kawasaki Disease: Updated AHA Guidelines. The Cardiology Advisor. Retrieved from: <https://www.thecardiologyadvisor.com/home/topics/pediatric-cardiology/kawasaki-disease-updated-aha-guidelines/>

RAAPID North
(for patients north of Red Deer)
(780) 735-0811

RAAPID South
(for patients in and south of Red Deer)
(403) 944-4486

References

1. American College of Rheumatology Clinical Guidance for Multisystem Inflammatory Syndrome in Children Associated With SARS-CoV-2 and Hyperinflammation in Pediatric COVID-19: Version 3. Henderson LA, Canna SW, Friedman KG, Gorelik M, Lapidus SK, Bassiri H, Behrens EM, Kernan KF, Schulert GS, Seo P, Son MBF, Tremoulet AH, VanderPluym C, Yeung RSM, Mudano AS, Turner AS, Karp DR, Mehta JJ. Arthritis Rheumatol. 2022 Apr;74(4):e1-e20. doi: 10.1002/art.42062. Epub 2022 Feb 3. PMID: 35118829; PMCID: PMC9011620
2. CDC website accessed on November 19th, 2022. Health Department-Reported Cases of Multisystem Inflammatory Syndrome in Children (MIS-C) in the United States. <https://covid.cdc.gov/covid-data-tracker/#mis-national-surveillance>
3. CDC website accessed on November 19th, 2022 16th. Information for Healthcare Providers about Multisystem Inflammatory Syndrome in Children (MIS-C). https://www.cdc.gov/mis/mis-c/hcp/index.html?CDC_AA_refVal=https%3A%2F%2Fwww.cdc